AN ERDŐS–RADO ORIENTED RAMSEY NUMBER DETERMINED: $k(3,4) = r(I_3, L_4) = 21$, BY EXPLICIT WITNESS AND CERTIFIED EXHAUSTION

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ABSTRACT. Let $k(n,m)$ be the least N such that every directed graph on N vertices contains an independent set of size n or a transitive tournament on m vertices. Determining $k(n,m)$ is a problem of Erdős and Rado (1967), #112 in the Erdős problems collection; it coincides with the oriented-graph Ramsey number $r(I_n, L_m)$ studied by Bermond and by Ihringer, Rajendraprasad, and Weinert (IRW). Beyond the classical tournament column $k(2,m)$, the known exact values were $k(3,3) = 9$ (Bermond 1974) and $k(4,3) = 15$, $k(5,3) = 23$ (IRW 2021); for $k(3,4)$, the case IRW single out as the next feasible one, the published bounds were $9 \le k(3,4) \le 25$. We determine $k(3,4) = r(I_3, L_4) = 21$.

The lower bound is an explicit oriented graph on 20 vertices, printed in full in §2 and checkable in milliseconds. The upper bound is an exhaustion of the 21-vertex case: a neighbourhood decomposition splits it into 346 propositional instances; CaDiCaL refuted every instance with an LRAT certificate, and every certificate was replayed by an independently written standard-library checker. Completeness of the decomposition was audited at content level: complete block inventories (three independent enumerations), a base encoding with no symmetry breaking whose models are exactly the $\{I_3, \mathrm{TT}_4\}$-free oriented graphs (brute-forced at small orders), and a one-to-one clause-multiset match of the 346 instance files against the decomposition re-derived from the definitions. With logically redundant confirming exhaustions at $N = 22, 23, 24$, the corpus is 445 certificates and just over 10^9 checked proof steps; certificates regenerate bit-for-bit from the recorded solver commands (8 of 8 attempts). Every component was then confirmed by an adversarial referee writing fresh code throughout. From the same campaign: $29 \le k(6,3) \le 33$, the lower bound new, by an explicit Cayley witness on 28 vertices; the upper bound is IRW's. We close with the questions the computation opens: the structure and possible uniqueness of the 20-vertex extremal graph, the $k(6,3)$ gap, the feasibility of $k(4,4)$ (the new value gives $k(4,4) \le 50$), and what a human-readable proof would require.

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Computation and authorship. All encodings, decompositions, searches, verifiers, and audits in this work were produced by **Claude** (Anthropic), directed by the author, on a single Apple M4 laptop. The independent proof checker imports only the Python standard library and shares no code with the solving pipeline; the adversarial referee of §4.2 was a separate agent instance given no access to the pipeline's code and wrote its own code throughout, including its own LRAT checker. This is a factual methods statement, and it is part of the point of the note: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result*. External tools used by the pipeline: CaDiCaL 3.0.1 (`--lrat`) and Python 3.

Prior-art record. The primary source [IRW21] was read in full on August 5, 2026 (arXiv v3; a copy is archived with the artifacts, and section references follow its numbering). The Erdős problems entry [Blo] was fetched live on August 5 and again on August 11, 2026 (including once more immediately before this draft was written): status OPEN, zero comments, zero claimed proofs, no exact value of any $k(n,m)$ recorded. An arXiv sweep of the same dates found no paper on $r(I_m, L_n)$ other than [IRW21], whose single recorded citation is off-target. The line has been dormant since 2020/2021. *Draft of August 11, 2026.*

1. Introduction

1.1. **The question.** An *oriented graph* is a loopless directed graph with at most one arc between any two vertices; a *tournament* is an oriented graph in which every pair is adjacent; a tournament is *transitive* when its arc relation is a linear order. Write I_n for the independent set on n vertices and L_m (equivalently TT_m) for the transitive tournament on m vertices. Following [IRW21], $r(I_n, L_m)$ is the least N such that every oriented graph on N vertices contains an independent set of size n or a transitive tournament on m vertices (for tournaments, subdigraph and induced-subdigraph containment coincide).

Erdős and Rado [ErRa67] asked for the same quantity over directed graphs: let $k(n, m)$ be minimal such that any directed graph on $k(n, m)$ vertices contains an independent set of size n or a transitive tournament of size m. The two functions are equal (Remark 1.3), and determining $k(n, m)$ is #112 in the Erdős problems collection [Blo], recorded there as open. That entry records the following. Erdős and Rado proved $k(n, m) \ll_m n^{m-1}$, with the explicit bound $k(n, m) \leq \big(2^{m-1}(n-1)^m + n - 2\big)/(2n-3)$; Larson and Mitchell [LaMi97] sharpened the dependence on m, in particular to $k(n, 3) \leq n^2$ — a bound also stated as [IRW21, Lemma 2.4], where it is attributed to [LaMi97]; and Z. Hunter observed $R(n, m) \leq k(n, m) \leq R(n, m, m)$, hence $k(n, m) \leq 3^{n+2m}$.

The exact values previously known form two families. The tournament column is classical: $k(2, 3) = 4$, $k(2, 4) = 8$, $k(2, 5) = 14$, $k(2, 6) = 28$ (see [IRW21] and the references there). In the third-column direction, Bermond [Ber74] proved $k(3, 3) = 9$; the extremal graph on 8 vertices is unique up to isomorphism by [IRW21, Lemma 3.1], which also gives its description as the circulant on $\mathbb{Z}_8$ with connection set $\{1, 6\}$. Ihringer, Rajendraprasad, and Weinert [IRW21] proved $k(4, 3) = 15$ and $k(5, 3) = 23$, gave the general bound $k(m, 3) \leq m^2 - m + 3$, and — combining their upper bound with Kim's lower bound for $r(I_m, K_3)$ — determined the order of growth $k(m, 3) = \Theta(m^2/\log m)$. For $k(3, 4)$ — the case [IRW21] single out in closing as the next one within reach — the record stood at

$$9 \;\leq\; k(3,4) \;\leq\; 25,$$

the lower bound by monotonicity from $k(3, 3) = 9$, the upper bound from the recursion of [IRW21, Lemma 2.3], $r(I_3, L_4) \leq 2\,r(I_3, L_3) + r(I_2, L_4) - 1 = 25$; the same value 25 is what [IRW21, Proposition 6.1], the small-parameter formula the authors describe as the state of the art for small m and n, returns at $(m, n) = (3, 4)$.

1.2. **Result.**

Theorem 1.1. $k(3, 4) = r(I_3, L_4) = 21$. *That is: every directed graph — equivalently, every oriented graph — on* 21 *vertices contains an independent set of size* 3 *or a transitive tournament on* 4 *vertices, and there is an explicit oriented graph on* 20 *vertices, exhibited in §2, containing neither.*

This determines the first previously unknown value in the $k(3, m)$ column of Erdős Problem #112. The proof is machine-checkable end to end. The lower bound is the explicit witness of §2. The upper bound is an exhaustive case decomposition of the 21-vertex problem into 346 propositional instances (§3), each refuted by CaDiCaL 3.0.1 [BFF+24] with an LRAT certificate [CFHKS17], in the spirit of cube-and-conquer [HKWB11] and of the certified exhaustions of [HKM16]; the completeness of the decomposition — that the 346 cases cover everything, with no symmetry breaking hidden in the encoding — was itself audited at content level (§4). Every certificate was verified by an independently written checker, and twelve of them by a second, unrelated one; the whole was confirmed by a final adversarial referee (§4.2).

Two secondary results from the same campaign are recorded at exact strength. The certified exhaustions at $N = 22, 23$ (and a structured instance at $N = 24$) are logically redundant given Theorem 1.1 but stand as independent confirmations (§5.1). And:

Proposition 1.2. $29 \le k(6,3) \le 33$. *The lower bound is new, by the explicit* 28*-vertex witness* $\mathrm{Cay}(\mathbb{Z}_{28}, \{3,8,10,12,17\})$*; the upper bound is* $m^2 - m + 3$ *at* $m = 6$*, due to* [IRW21].

$k(6,3)$ remains open, and nothing stronger about it is claimed here.

Remark 1.3 (Directed versus oriented graphs). Erdős Problem #112 is phrased for directed graphs, which may carry 2-cycles; the computation here is over oriented graphs. The two Ramsey functions coincide. Every oriented graph is a directed graph, so $k(n,m) \ge r(I_n, L_m)$. Conversely, a directed graph avoiding both patterns yields an oriented one on the same vertex set: delete one arc of each 2-cycle. Deletion preserves adjacency and non-adjacency of every pair — hence preserves every independent set — and only removes transitive subtournaments; so $k(n,m) \le r(I_n, L_m)$. The extremal orders are equal, and the value 21 answers the problem as posed.

1.3. **What is not claimed.** We do not claim to resolve Erdős Problem #112. The problem asks for $k(n,m)$ as a function of two parameters and is not settled by any finite computation; a single new entry in its table leaves it open. We do not claim a human-readable proof of the upper bound: what a skeptic must believe and can replay is stated in §7; what a human proof would require, in §8. We do not claim uniqueness of the 20-vertex extremal graph (Question 8.1). The asymptotics of $k(n,m)$ are untouched: the best general results remain those of [ErRa67, LaMi97, IRW21]. We do not claim $k(6,3)$: Proposition 1.2 narrows its range and no more. The prior bounds improved on here are credited at their sources: the interval $[9, 25]$ is [Ber74] plus [IRW21].

2. The lower bound: an explicit graph on 20 vertices

Table 1 defines an oriented graph W on vertex set $\{0, \dots, 19\}$ by its out-neighbourhoods; W has 126 arcs.

$0 \to 1,3,4,7,9,18$	$10 \to 2,3,5,6,17,19$
$1 \to 2,4,6,7,10,14,16$	$11 \to 2,4,6,8,9,10$
$2 \to 0,8,12,16,17,18,19$	$12 \to 9,10,11,15,17,19$
$3 \to 1,6,11,12,13,16$	$13 \to 0,1,9,10,11,14,17$
$4 \to 3,5,10,12,13,14$	$14 \to 0,3,6,16,18,19$
$5 \to 2,6,11,12,13,15,18$	$15 \to 0,1,2,8,9,14$
$6 \to 4,8,12,13,17,19$	$16 \to 0,4,7,13,15,17$
$7 \to 2,3,13,14,15,18$	$17 \to 1,3,4,5,8,11$
$8 \to 3,5,7,9,10,12,18$	$18 \to 1,9,13,15,16,19$
$9 \to 2,5,7,14,16,19$	$19 \to 0,4,5,7,8,11,15$

Table 1. The 20-vertex witness W: out-neighbourhoods.

Theorem 2.1. *W contains no independent set of size* 3 *and no transitive tournament on* 4 *vertices. Hence* $k(3,4) \ge 21$.

The check is a finite verification over the $\binom{20}{3} = 1140$ triples and $\binom{20}{4} = 4845$ quadruples, and runs in milliseconds; it was performed by independently written verifiers using three distinct transitivity criteria (among them the score-sequence criterion: transitive if and only if the scores are $\{0,1,2,3\}$), and again from scratch by the adversarial referee of §4.2, who additionally pushed a randomly relabelled copy of W through the full canonicalization of §3.3 at every one of its 20 vertices.

Three facts about W, directly checkable from Table 1, are worth recording for §8. Every out- and in-degree is 6 or 7, and the vertex types $\big(d^+(v), d^-(v), |I(v)|\big)$ — out-degree, in-degree, number of non-neighbours — take exactly four values: $(6,6,7)$ at eleven vertices, and $(6,7,6)$, $(7,6,6)$, $(7,7,5)$ at three vertices each. In particular W is not vertex-transitive. Finally, at each vertex of type $(6,6,7)$ the non-neighbourhood is a 7-vertex TT_4-free tournament, hence (Proposition 3.3) a copy of the Paley tournament QR_7.

3. The upper bound: exhaustion at $N = 21$

Call an oriented graph *free* if it contains neither I_3 nor TT_4. The upper bound is the assertion that no free graph on 21 vertices exists. The exhaustion has three mathematical ingredients — a neighbourhood lemma, two vanishing endpoints, and complete inventories of the neighbourhood blocks — and one computational engine: 346 certified refutations.

3.1. **The case decomposition.** For a vertex w of an oriented graph D, write $N^+(w)$, $N^-(w)$, $I(w)$ for the sets of out-neighbours, in-neighbours, and non-neighbours of w.

Lemma 3.1. *Let D be free and w a vertex of D. Then (i) $N^+(w)$ and $N^-(w)$ each induce an oriented graph with no I_3 and no TT_3; (ii) $I(w)$ induces a tournament with no TT_4.*

Proof. (i) An independent triple inside $N^+(w)$ is one in D. A transitive triangle inside $N^+(w)$ extends by w, a common in-neighbour of its three vertices, to a TT_4 in D. The argument for $N^-(w)$ is the same with w as common out-neighbour (sink side). (ii) Two non-adjacent vertices of $I(w)$ together with w form an I_3, so $I(w)$ induces a tournament; a TT_4 inside it is one in D. □

Lemma 3.2. *There is no $\{I_3, \mathrm{TT}_3\}$-free oriented graph on 9 vertices, and no TT_4-free tournament on 8 vertices.*

Lemma 3.2 is equivalent to $k(3,3) = 9$ [Ber74] and to the classical $k(2,4) = 8$, but the exhaustion does not rest on the literature for it: both statements were established internally four ways — by direct enumeration of all labelled graphs of the two families (the labelled counts of $\{I_3, \mathrm{TT}_3\}$-free graphs on $1, \ldots, 9$ vertices are 1, 3, 20, 224, 2554, 18370, 30960, 5040, 0; of TT_4-free tournaments on $1, \ldots, 8$ vertices, 1, 2, 8, 40, 184, 240, 240, 0), re-derived twice more by independent enumerations during the audits, and by LRAT-certified refutations of the corresponding instances, replayed by both checkers.

By Lemma 3.1 and Lemma 3.2, in a free graph every vertex w has $d^+(w) \le 8$, $d^-(w) \le 8$, $|I(w)| \le 7$; counting $N = 1 + d^+ + d^- + |I| \le 24$ caps free graphs at 24 vertices, hence $k(3,4) \le 25$ by pure arithmetic. This is the specialization to $(3,4)$ of the argument of [IRW21, Lemma 2.3], and it is exactly where the previous record stood. The exhaustion works at $N = 21$: there $(p,q,s) := (d^+(w), d^-(w), |I(w)|)$ satisfies $p + q + s = 20$ with $p, q \le 8$, $s \le 7$, which also forces $p, q \ge 5$ and $s \ge 4$.

Proposition 3.3 (Block inventories)**.** *Up to isomorphism, the $\{I_3, \mathrm{TT}_3\}$-free oriented graphs on $5, 6, 7, 8$ vertices number 25, 31, 7, 1 respectively — the unique one on 8 vertices being Bermond's circulant, in agreement with* [IRW21, Lemma 3.1] *— and the TT_4-free tournaments on $4, 5, 6, 7$ vertices number 3, 3, 1, 1, the unique one on 7 vertices being the Paley tournament QR_7.*

These inventories were computed by orbit-peeling of the labelled enumerations under S_k, with a built-in closure assertion (every relabelled image of every representative must reappear in the labelled set), and re-derived by three independent enumerations in total; the labelled orbit sizes cross-check exactly against the automorphism groups (order 8 for Bermond's graph, 21 for QR_7).

Reversing all arcs preserves non-adjacency (hence I_3-freeness), maps TT_4 to TT_4, and swaps $(p,q,s) \mapsto (q,p,s)$; both block families of Proposition 3.3 are closed under reversal. So one may assume $p \le q$, and the possible types at $N = 21$ are exactly six. Assigning to each type every ordered choice of isomorphism classes for the three blocks (both ordered pairs are kept when $p = q$) yields the case list of Table 2: 346 cases in all.

type (p,q,s)	block classes	cases
$(5,8,7)$	$25 \times 1 \times 1$	25
$(6,7,7)$	$31 \times 7 \times 1$	217
$(6,8,6)$	$31 \times 1 \times 1$	31
$(7,7,6)$	$7 \times 7 \times 1$	49
$(7,8,5)$	$7 \times 1 \times 3$	21
$(8,8,4)$	$1 \times 1 \times 3$	3
	total	346

TABLE 2. The six vertex types at $N = 21$ (with $p \leq q$) and the case count.

3.2. **The encoding.** The base formula base(N) has one Boolean variable per ordered pair ($x_{u\to v}$: the arc $u \to v$ is present; $2\binom{N}{2} = 420$ variables at $N = 21$) and three clause families, transcribing the definition and nothing else: for each pair, no 2-cycle ($\neg x_{u\to v} \vee \neg x_{v\to u}$); for each triple, not independent (the disjunction of its six arc variables); for each 4-set, for each of the 24 transitive orientations, a clause forbidding that orientation. At $N = 21$ this is $\binom{21}{2} + \binom{21}{3} + 24\binom{21}{4} = 145{,}180$ clauses. There is no symmetry breaking of any kind in base(N): its models correspond to free oriented graphs and to nothing else, and this semantic claim was verified by brute force at small orders — over all $3^{\binom{N}{2}}$ orientations, the models of base(4) are exactly the 612 free graphs among 729, and the models of base(5) exactly the 35,488 among 59,049.

Each of the 346 cases becomes a *cube*: base(21) together with unit clauses fixing vertex 0's complete arc pattern (arcs to $1,\dots,p$, arcs from $p+1,\dots,p+q$, non-adjacency to the rest) and fixing each of the three blocks to its chosen class representative — $2\big(20 + \binom{p}{2} + \binom{q}{2} + \binom{s}{2}\big)$ units in all, two per constrained pair. All arcs between different blocks are left entirely free, constrained only by base(21).

3.3. **Soundness of the case split.**

Proposition 3.4. *Suppose some free oriented graph D on 21 vertices existed. Then, for one of the 346 cubes, some relabelling of D (of D with all arcs reversed, if necessary) would satisfy every clause of that cube.*

Proof. Fix any vertex w of D and let (p,q,s) be its type; by Lemma 3.1, Lemma 3.2, and $p+q+s = 20$, the type appears in Table 2 after reversing all arcs if $p > q$ (reversal preserves freeness and swaps the type; when $p = q$ both ordered block pairs are among the cubes, so no choice is lost). By Lemma 3.1 the three blocks induced by $N^+(w)$, $N^-(w)$, $I(w)$ belong to the enumerated families, and by the completeness of the inventories (Proposition 3.3) each is isomorphic to exactly one class representative. Relabel $w \mapsto 0$, $N^+(w)$ onto $\{1,\dots,p\}$, $N^-(w)$ onto $\{p+1,\dots,p+q\}$, and $I(w)$ onto the rest, choosing within each block an isomorphism onto its representative. The relabelled graph is free, so it satisfies base(21) (whose models are exactly the free graphs), and by construction it satisfies every unit clause of the cube indexed by its type and block classes. □

Since all 346 cubes are unsatisfiable — §3.4 — no free graph on 21 vertices exists, and with Theorem 2.1, Theorem 1.1 follows. The proof above is a short argument whose computational inputs are Lemma 3.2, the completeness of Proposition 3.3, the semantics of base(N), the identity of the 346 cube files with base(21)-plus-units, and the 346 refutations; every input was machine-checked independently, most several times over (§4).

3.4. **The certificates.** CaDiCaL 3.0.1 [BFF+24] refuted each of the 346 cubes, emitting an LRAT proof [CFHKS17] for each; every proof was verified by an independently written checker that imports only the Python standard library, accepts only on derivation of the empty clause, and rejects deliberately corrupted controls. The verification ledger records, per certificate, the verdict, the checked-step count, the SHA-256 digests of formula and of proof, and the proof's byte count. Across the $N = 21$ layer the certificates range from 236,431 to 7,816,216 checked steps, about 245 GB of uncompressed proof in total; the complete corpus of the campaign (including the redundant layers of §5.1) is 445 certificates and just over 10^9 checked steps. All 346 solver logs report unsatisfiability; every base has exactly one ledger line, verdict VERIFIED, formula digest matching the audited file; no failures, no rejections.

4. Verification and audit

4.1. **Completeness at content level.** The decomposition was audited by a referee-style completeness audit, with independent code, and later re-derived once more from scratch: the required cube inventory was recomputed from the type arithmetic and fresh block enumerations, and each of the 346 cube files was required to equal base(21) plus its expected units *as a clause multiset* — no missing clause, no duplicate, no stray unit, exact DIMACS header — with the match established one-to-one across the whole layer, filename-blind. Negative controls (a dropped clause, a duplicated clause, a hidden extra unit, a flipped literal) were all rejected. The same audit passed for the $N = 22$, $N = 23$, and $N = 24$ layers.

As a positive control, the identical pipeline — same generator, same unit-fixing — was run at $N = 20$, where a free graph is known to exist: the witness W, canonicalized per Proposition 3.4 onto its class representatives, satisfies all 117,750 clauses of the generated $N = 20$ cube; the solver reports satisfiability on both the cube and the pure instance; and the extracted models were independently re-verified free. A false-unsatisfiability pipeline would have failed this control.

Two operational events are recorded for honesty. Mid-campaign the checker's hint semantics were relaxed from strict format checking to sound fixpoint semantics (skip hints unusable under the current assignment, hard-fail on missing identifiers, accept only on the empty clause) to accommodate a solver emission quirk; the change was ruled sound, and all negative controls still fail. And three ledger lines were poisoned by infrastructure failures and repaired: one when the checker process was killed without output under memory and disk pressure, and later two more (a second such kill, and a cloud-storage file eviction). In every case only the verdict field changed — never a formula and never a proof. The two later originals survive in a ledger backup whose formula and proof digests and byte counts are identical to the current lines; the first survives verbatim in the failure log, the backup having been taken afterwards. All three certificates verify with matching digests, two of them under the referee's independent checker and one regenerated byte-identically from its recorded solver command.

4.2. **The adversarial referee.** The confirmed result rests on a final adversarial referee pass, by a separate agent instance under an explicit independence rule: every load-bearing check used code written fresh in that session, sharing nothing with the pipeline or the earlier audits beyond the DIMACS variable-numbering convention. The referee re-derived the entire skeleton from the definitions — the neighbourhood lemma, both endpoints of Lemma 3.2, the inventories of Proposition 3.3, the six types, the count 346 — and re-established the content bijection between the 346 required formulas and the files on disk over regenerated content. It wrote and control-validated its own LRAT checker, then attacked ten adversarially chosen certificates: the largest proof in the corpus, all three entries with a repair history (these four are the 2.0–2.3 GB proofs, four of the five largest), the smallest, the largest of every remaining type, and one further cube. All ten verified, with step counts and digests equal to the ledger's in every field; in all twelve cases where both checkers ran the same certificate, their step counts agreed exactly. Eight of the ten proofs were regenerated from nothing but the formula by re-running the recorded solver command; all eight came out byte-identical to the recorded certificates. The canonicalization of Proposition 3.4

was attacked with 57 pivot canonicalizations across three graphs, including a family member at 18 vertices found by simulated annealing from a uniformly random start, sharing no ancestry with the witness; no escape from the inventory was found. Novelty was swept live the same day. On this protocol the result was confirmed on August 11, 2026; the referee's verdict record and fresh code ship with the artifacts.

5. Redundant layers, and $k(6,3)$

5.1. **Certified exhaustions at $N = 22, 23, 24$.** Freeness is hereditary, so the $N = 21$ exhaustion already implies that no free graph exists on any larger order. The campaign nevertheless produced certified exhaustions at $N = 23$ (8 cubes: types $(7,8,7)$ across the seven 7-vertex block classes, and $(8,8,6)$) and $N = 22$ (90 cubes: types $(6,8,7)\times 31$, $(7,7,7)\times 49$, $(7,8,6)\times 7$, $(8,8,5)\times 3$), as well as a single structured instance at $N = 24$; all 99 certificates are verified and ledgered, and the multiset completeness audit of §4.1 passed for each layer. These layers are logically redundant given Theorem 1.1; they stand as independent confirmations along the descent $25 \to 24 \to 23 \to 22 \to 21$, and their certificates are part of the corpus.

5.2. **The bracket for $k(6,3)$.**

Proof of Proposition 1.2. Upper bound: $k(m,3) \leq m^2 - m + 3$ is [IRW21, Proposition 3.4]; at $m = 6$ this is 33. Lower bound: let $S = \{3, 8, 10, 12, 17\} \subset \mathbb{Z}_{28}$ and let $D = \mathrm{Cay}(\mathbb{Z}_{28}, S)$, with arcs $x \to x + s$ for $s \in S$. Since $-S = \{11, 16, 18, 20, 25\}$ is disjoint from S, D is an oriented graph; it is 5-regular in and out, with 140 arcs. A transitive triangle $x \to y \to z$, $x \to z$ in $\mathrm{Cay}(G, S)$ is exactly a pair $a = y - x$, $b = z - y$ in S with $a + b = z - x$ also in S; here $S + S = \{1, 6, 11, 13, 15, 16, 18, 20, 22, 24, 25, 27\}$ is disjoint from S, so D has no TT_3. The underlying graph of D is $\mathrm{Cay}(\mathbb{Z}_{28}, S \cup -S)$, whose independence number is 5; so D has no I_6. Hence $k(6,3) \geq 29$. □

Only the independence number is a machine check, over the $\binom{28}{6} = 376{,}740$ six-subsets. It was performed by the independently written witness verifiers of §2, and again by exact computation of the independence number by branch and bound. Two further 28-vertex witnesses, one of them a Cayley digraph over $\mathbb{Z}_{14} \times \mathbb{Z}_2$, verify equally and are archived.

Disjoint unions cannot reach 28: splitting the independence budget 5 across components with the extremal component orders available (3 at budget 1, since a TT_3-free tournament has at most 3 vertices; $k(a+1, 3) - 1$ in general) gives $3 + 22 = 25$ as the best partition, so every witness on 26 or more vertices — including D, whose connection set contains a generator — is connected in the underlying adjacency. No lower bound beyond 24 (W_{22}, the $\{I_5, \mathrm{TT}_3\}$-free circulant of [IRW21, Observation 4.2], plus an isolated vertex) appears in the literature.

6. Certification

The permanent record of the computation is deliberately small: the verification ledger (one line per certificate: verdict, checked-step count, SHA-256 of formula and of proof, and the proof's byte count), the queue files recording the exact solver command line for every cube of the $N = 21$, 22 and 23 layers, the generator and checker scripts, the block-inventory representatives the generator consumes, the witness files, and the audit and referee records. All of it is deposited with this note in the program's public certificate repository *Certify* (`https://github.com/05oz/certify`; concept DOI 10.5281/zenodo.21799111) at release, under a version DOI minted then. The multi-gigabyte proofs themselves are a cache, not the record: each regenerates from its recorded command — deterministically, in our environment: the referee's eight regenerations were all byte-identical to the originals, matching the ledger's digests — and the repository documents the regeneration procedure. To replay: verifying any shipped or regenerated certificate, the witnesses, the enumerations, and the content-level completeness audit requires only CPython (the checkers are standard-library only); regenerating proofs requires CaDiCaL 3.0.1.

7. The trusted base

What a skeptic must believe, separated from what they can replay.

Machine-checked, replayable with standard-library Python: the witness W and the 28-vertex witness of Proposition 1.2 (milliseconds); the labelled enumerations behind Lemma 3.2 and Proposition 3.3, performed three independent times, with closure assertions (minutes); the brute-force semantics of base(4) and base(5) over all orientations; the clause-multiset identity of all 445 cube files with their re-derived specifications; and the LRAT replay of all 445 certificates by an independently written checker, with twelve of them re-run by a second, unrelated checker agreeing on the exact step count in every case. Both checkers reject all negative controls.

Trusted:

(1) *The checkers' proof semantics.* Both checkers implement RUP-with-hints with sound fixpoint treatment of hints and accept only on derivation of the empty clause. That this discipline is sound is a short argument on paper, not a machine-checked one; the mitigation is two independent implementations in agreement, plus negative controls (corrupted proof, mutated formula, dropped hint) rejected by both.

(2) *The assembly of §3.3.* The proof of Proposition 3.4 is half a page of mathematics to be read by a human. Every computational input to it is machine-checked; the gluing is not, and is short by design.

(3) *The generalization of the base semantics from $N = 4, 5$ to $N = 21$.* The brute-force model check is exhaustive only at $N = 4, 5$; at $N = 21$ the claim rests on the clause families being uniform in N (with closed-form clause count $\binom{N}{2} + \binom{N}{3} + 24\binom{N}{4}$ matching at every audited order) and on the independent re-derivation of the exact clause multiset from the definitions.

(4) *SHA-256 collision resistance*, for the bookkeeping that ties ledger lines to files — not for the mathematics, since a skeptic can regenerate and re-check everything without trusting any hash.

(5) *CPython, the operating system, and the hardware.*

CaDiCaL, the cube generators, and the solving pipeline are *not* in the trusted base: nothing depends on the solver being correct, only on its proofs checking, and every proof was checked by code unrelated to the solver — a sample of them by two such checkers independently.

8. Questions

What is this computation *for*? Four directions it opens.

8.1. The extremal structure at $(3, 4)$. The extremal graph for $k(3, 3)$ is unique up to isomorphism [IRW21, Lemma 3.1] — a circulant. The witness W of §2 shows no such symmetry: it has four vertex types, so it is not vertex-transitive, although the Paley tournament QR_7 appears as the non-neighbourhood of eleven of its twenty vertices. This mirrors the situation at $k(4, 3)$, where [IRW21] remark that no Cayley witness on 14 vertices exists, while the $k(5, 3)$ witness is again circulant.

Question 8.1. Is W the unique $\{I_3, \mathrm{TT}_4\}$-free oriented graph on 20 vertices up to isomorphism? If not, what is the number of extremal graphs, and does any admit an algebraic description?

A related quantitative puzzle: the counting bound of §3.1 caps free graphs at 24 vertices, yet the largest free graph has 20 — at $N = 21, 22, 23$ every one of the $346 + 90 + 8$ cases dies, with no tightness anywhere. A structural explanation of this slack of 4 (at $k(3, 3)$ the analogous cap

$1+2+2+3 = 8$ is attained, by Bermond's graph) is exactly what a human proof would have to supply.

8.2. **The gap at $k(6,3)$.** Proposition 1.2 leaves $k(6,3) \in [29,33]$, now the smallest open case of the $k(m,3)$ column. Every witness on 26 or more vertices is connected in the underlying adjacency, which points toward vertex-transitive candidates: for Cayley digraphs, TT_3-freeness of $\mathrm{Cay}(G,S)$ is exactly sum-freeness of S, which makes the search cheap. This is how the 28-vertex witness was found. An exhaustive sweep of connection sets over the cyclic groups of orders 29, 30 and 31 — the only abelian groups of those orders — and over three of the seven abelian groups of order 32 produced nothing, so any witness beating 28 lies outside that range of constructions. The upper bound looks harder: an exhaustion in this note's style would need complete inventories of $\{I_5, \mathrm{TT}_3\}$-free graphs on up to 22 vertices, far beyond present enumeration, so progress below 33 seems likelier to come from sharpening the $m^2 - m + 3$ argument of [IRW21].

8.3. **$k(4,4)$ and beyond.** The new value propagates through the standard recursion. By Lemma 3.1 in its general form [IRW21, Lemma 2.1], every $\{I_4, \mathrm{TT}_4\}$-free graph has $d^{\pm}(v) \leq k(4,3) - 1 = 14$ and $|I(v)| \leq k(3,4) - 1 = 20$, so such a graph has at most $1 + 14 + 14 + 20 = 49$ vertices, and likewise every $\{I_3, \mathrm{TT}_5\}$-free graph has at most $1 + 20 + 20 + 13 = 54$; hence

$$k(4,4) \;\leq\; 50, \qquad\qquad k(3,5) \;\leq\; 55,$$

which is the recursion of [IRW21, Lemma 2.3] evaluated at the new value. Meanwhile $k(4,4) \geq k(3,4) = 21$, since a graph with no I_3 has no I_4, so W is also $\{I_4, \mathrm{TT}_4\}$-free. An honest feasibility assessment: determining $k(4,4)$ or $k(3,5)$ by the present method is out of reach. The decomposition would require complete isomorph inventories of $\{I_4, \mathrm{TT}_3\}$-free graphs up to 14 vertices, or of $\{I_3, \mathrm{TT}_4\}$-free graphs up to 20 vertices — and the labelled counts of the latter family are already 35,488 at 5 vertices and 4,490,572 at 6 — and the resulting instances at N in the forties would dwarf the 10^9 checked steps spent here at $N = 21$. What does look feasible: raising lower bounds by witness search (any verified witness is a one-line check), and isomorph-free generation of the free families at small orders to map the terrain.

8.4. **Toward a human-readable proof.** The upper-bound half of Theorem 1.1 is 346 machine proofs averaging about two and a half million checked steps. Two concrete paths would shrink it. First, structure theory: the proof of Proposition 3.4 shows that any free graph on 21 vertices is assembled from three blocks drawn from very short lists around every vertex simultaneously; a human argument that such an assembly forces an I_3 or a TT_4 — perhaps via the QR_7 and Bermond blocks, which dominate the type table — would replace the certificates outright, and would likely explain the slack of §8.1 as a by-product. Second, proof compression: the LRAT corpus is an upper bound on the difficulty of the theorem, not a measure of it, and nothing is known about the minimum certificate size; a decomposition chosen for proof economy rather than for engineering convenience might cut the corpus by orders of magnitude and expose which cases are genuinely hard. Either advance would turn a determined value into an understood one.

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the adversarial referee protocol of §4.2 and the referee's rule that a computation-only paper close with the questions it opens shaped this note's final form.

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